ResQVision AI: Technical Architecture

Battle Damage Assessment & Disaster Response

1. Executive Summary

ResQVision AI is a specialized, end-to-end Machine Learning pipeline engineered for automated structural damage assessment using pre- and post-disaster satellite imagery. Designed primarily for high-stakes operational environments, the system translates seamlessly into military Battle Damage Assessment (BDA) by identifying rapid structural destruction caused by heavy weaponry or natural events.

By automating high-resolution imagery inspection, the system drastically reduces cognitive load on tactical analysts. This report details the two-stage computer vision architecture, the dataset preparation, the hyperparameter configurations, and the interactive intelligence dashboard deployed for operational decision-making.

2. Model Architecture: Two-Stage Siamese Network

Direct pixel-level damage classification across raw imagery is error-prone because deep neural networks frequently confuse background environmental shifts with targeted structural degradation. ResQVision AI resolves this using a modular Two-Stage Pipeline.

Stage 1: Building Footprint Localization

The first stage isolates and registers building footprints using strictly the pre-event image. It deploys a U-Net architecture featuring an ImageNet-pretrained ResNet34 encoder. The network outputs a calibrated binary mask distinguishing background (0) from structure (1).

Stage 2: Damage Severity Classification

The second stage evaluates structural degradation using a Twin-Tower Substrate (TTS) Siamese network. Pre- and post-event images are processed through shared encoder weights. The architecture computes element-wise feature differences (|post - pre|) at each scale to expose localized structural alterations, producing a 5-class severity map: Background, No Damage, Minor Damage, Major Damage, and Destroyed.

3. Dataset Configuration & Training Protocol

The project is trained on the xView2 (xBD) dataset. To respect RTX 2050 4GB VRAM limits, satellite tiles are resized via Bilinear interpolation to 512x512. The data partition maintains an 80% training split (800 image pairs) and a 20% validation split (200 image pairs).

Key Hyperparameters:

- **Training Epochs:** 5 complete optimization cycles.
- **Batch Size:** 4 paired image tiles per iteration.
- **Optimizer:** AdamW with learning rate 1e-4.
- **Loss Function:** Weighted CrossEntropyLoss.

To mitigate heavy class imbalance where undamaged structures dominate, class loss weights are scaled to heavily penalize errors on destroyed objects: Background (0.2), No Damage (1.0), Minor (2.0), Major (3.0), and Destroyed (4.0).

4. Tactical Intelligence Dashboard

The dual model checkpoints (localization_best.pth and damage_classifier_best.pth) are integrated into a streamlined Streamlit application designed for rapid field deployment.

Integrated Functional Modules:

- **Dual-Pane Slider:** Interactive split-view slider for side-by-side pre- and post-event satellite inspection.
- **Dynamic Analytics:** Plotly donut visualizations aggregating localized structural damage metrics.
- **Threshold-Based Alerts:** Automatic classification of target zones into Critical, Elevated, or Stable threat levels.
- **Data Export:** Standardized CSV export for downstream mission intelligence integration.

5. Quantitative Evaluation & Conclusion

To establish academic and operational validity, evaluation scripts generate standard computer vision benchmarks: Intersection over Union (IoU) for Stage 1 localization and Macro-averaged F1-Score across the discrete damage categories.

Conclusion

ResQVision AI establishes an end-to-end framework translating complex remote sensing data into practical, field-grade intelligence. Combining a Twin-Tower Siamese backbone with an accessible analytical dashboard produces an effective, production-ready solution for Battle Damage Assessment and disaster mitigation workflows.